Inference Problems

1. Standard error $= \frac{10}{\sqrt{100}} = 1$

Test statistic $= \frac{52-50}{1} = 2$

α	$Z >$ critical?		
.1	$1.645 < 2$	yes	reject H_0
.05	$1.96 < 2$	yes	Reject H_0
.01	$2.576 > 2$	no	fail to reject

95% CI

$52 \pm 1.96 = (50.04, 53.96)$

2. $Z = \frac{3.1-0}{1.3} = 2.38$

α	Margin	CI	Reject?
.1	2.139	.96, 5.24	Reject
.05	2.548	.55, 5.65	Reject
.01	3.349	-.25, 6.45	Fail to Reject

3. $SE = \frac{3}{\sqrt{36}} = .5$

95% CI $11.5 \pm 1.96(.5) = (10.52, 12.48)$

$Z = \frac{11.5-12}{.5} = -1 \quad |-1| < 1.96$

Fail to reject the null at .005.

12 is inside the confidence Interval.

4. $SE = \sqrt{\frac{64}{100} + \frac{49}{100}} = 1.063$

$Z = \frac{25-22}{1.063} = 2.82$

95% CI $3 \pm 1.96(1.063) = (0.92, 5.08)$

The interval does not include 0 so reject the null at $\alpha = .05$. This means $\mu_A \neq \mu_B$